



*Software Engineering (3) Course (SWE521): Agile
Software Engineering Skills*

AGILE SOFTWARE ENGINEERING SKILLS

LECTURE 02

Self-Organizing Teams Exercises

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Introduction

- Exercises have focused on developing an understanding of the skills available in your team, developing shared goals to improve productivity and establishing good behavioral practices within your team.
- In Chap. 3, I will look at the other roles that are defined when using agile software development methods. These roles are important, for supporting self-organizing teams that develop software.



Exercise 2.1 (Review The Material)

- **The first exercise is to review the material in each chapter.**
 - Write a few notes in your learning journal for this chapter.
 - These could be brief notes, just a few bullet points. Or perhaps you want to create a longer essay.
 - **Hint:**
 - In this chapter, we explored
 - Self-organizing teams
 - Agile principles
 - Forming teams
 - Collaboration activities within self-organizing teams
 - Virtual teams
- You should aim to write down a few comments about the things you learned in each topic.



Exercise 2.2 (Identify the various skills) 1 to 9

Work with the other members of your team to identify the various skills you have between you.

As a group, you should discuss two main areas:

1. What are the types of skills and skill categories available within the group?
2. What metrics or experience levels do you use to assess a skill competency?

You will want to include on technical skills in your inventory.

1. What technologies have you worked with?
2. What techniques have you learned (across the development lifecycle, perhaps)?
3. What do you know already about agile methods?

Some people might find the following categorization useful:

1. Software
2. Hardware
3. Networks
4. Information management or storage
5. Processes



Exercise 2.2 (Identify the various skills) 2 to 9

Don't forget to include non-technical skills.

Think about your hobbies, interests and pastimes. What skills have you acquired during these other activities?

These might loosely be called **soft skills** (administrative, organizational, social and communication skills, etc.).

You might have all sorts of skills that will be useful to your group.

It is best not to be shy about the range of skills you have.

Something you take for granted might be seen as a huge asset to another member of the group.

In terms of experience levels, you might find the following categorization useful:

1. Novice (I was taught a University course on this and completed some assignments.)
2. Competent (I've been using this routinely for a couple of years.)
3. Proficient (I have 3–5 years' experience with this.)

Create a spreadsheet or use some other way to collate and capture all the skills in your group.

If your group is rather homogeneous, then you might have a lot of similar skills.

your group is more heterogeneous, coming from a wide range of different cultures, experiences and previous backgrounds.

A heterogeneous group will likely have a wider range of different skills that you can draw on.



Exercise 2.2 (Identify the various skills) 3 to 9

- **Hint**

There are, at least, three aspects to this exercise.

1. What **skills** do you have?
2. What **level of expertise** do you have in each skill?
3. how can we **guide our own professional development** in the future?

We need some categories of skills and skill levels.

Learning about this will help us identify our own strengths.

We can then use the exercise to focus our own professional development and training activities.

The **Dreyfus** model of skills acquisition has five levels: **novice**, **advanced beginner**, **competent**, **proficient** and **expert**.

1. A **novice** needs close **supervision**.
2. An **advanced beginner** is able to achieve some steps using **own judgement**.
3. Someone **competent** is able to achieve most **steps using own judgement**.
4. Someone **proficient** is able to **take full responsibility** for own work and coach others.
5. An **expert** is able to go **beyond existing standards**.

This simple model is probably sufficient for this exercise.



Exercise 2.2 (Identify the various skills) 4 to 9

in Skills Framework for the Information Age (SFIA), there are seven skill levels .

SFIA identifies level of experience ranging from entry–level professional through to industry–wide thought leader.

The **highest level** is aimed at **senior executives** (such as the Chief Technology Officer) of large corporations or Government agencies.

These levels are:

1. Follow.
2. Assist.
3. Apply.
4. Enable.
5. Ensure and advise.
6. Initiate and influence.
7. Set strategy, inspire and mobilize.



Exercise 2.2 (Identify the various skills) 5 to 9

- The levels of responsibility are described in terms of five generic attributes:
- autonomy, influence, complexity, knowledge and business skills.
- SFIA provides a detailed breakdown of expectations at each skills level, developed over many years.
- Consequently, SFIA provides lots of useful information about how to improve your skills.
- **What about the skill areas themselves?**
- I mentioned that you might want to consider six categories:
 1. Software
 2. Hardware
 3. Networks
 4. Information management or storage
 5. Processes
 6. Soft (transferable) skills



Exercise 2.2 (Identify the various skills) 6 to 9

But, this might be too tied to specific technologies and not sufficiently focused on the objectives you are trying to achieve.

SFIA also has six categories of professional skills, but they are more focused on the goal or purpose:

- Strategy and architecture
- Change and transformation
- Development and implementation
- Delivery and operation
- Skills and quality
- Relationships and engagement



Exercise 2.2 (Identify the various skills) 7 to 9

These categories are divided into sub-categories and around 100 different skill areas. For example, the development and implementation category includes:

- Systems development
 - Systems development management
 - Software design
 - Software development
 - Database design
 - Network design
 - Testing
- User experience
 - User experience analysis
 - User experience design
 - User experience evaluation
- Installation and integration
 - Systems integration and build
 - software configuration
 - Systems installation



Exercise 2.2 (Identify the various skills) 8 to 9

- SFIA used to learn about how to develop new aspects of your skill set.
- For each **category**, there is a breakdown (details) of how the skill is applied at the different levels.
- For example, in the **Systems development category** and **Programming/software development** sub-category:
 - **Level 2** is described as ‘Designs, codes, verifies, tests, documents, amends and refactors simple programs/scripts.
Applies agreed standards and tools, to achieve a well-engineered result. Reviews own work’.
 - **Level 3** ‘Designs, codes, verifies, tests, documents, amends and refactors **moderately complex** programs/scripts.
Applies agreed standards and tools, to achieve a well-engineered result.
Collaborates in reviews of work with others as appropriate’.



Exercise 2.2 (Identify the various skills) 9 to 9

- **Level 4** ‘Designs, codes, verifies, tests, documents, amends and refactors complex programs/scripts and **integration software services**.

Contributes to selection of the software development approach for projects, selecting appropriately from **predictive (plan-driven)** approaches or **adaptive (iterative/ agile)** approaches.

Applies agreed standards and tools, to achieve well-engineered outcomes. Participates in reviews of own work and **leads reviews of colleagues’ work**.

As you can see from the above example, the levels in SFIA illustrate a growth in experience and influence.

The SFIA detailed matrix of skills can also be used to create job descriptions as well as support the personal and professional development of practitioners.



Exercise 2.3 (Personal Learning Timeline) 1 to 3

A personal learning time line is a graph of your life so far.

Time goes on the x-axis of your graph.

The y-axis represents your learning.

- **High values** are for **periods** in your life when you **learned a lot**.
- **Low values** are when **you did not learn** so much.

Best to use a large sheet of paper, such as a sheet of flip chart paper.

This exercise provides a vehicle (**chance**) for learning about each other's life story.

You will probably **learn** some **surprising things** about friends and colleagues.



Exercise 2.3 (Personal Learning Timeline) 2 to 3

This exercise can help you bond (communicate) **with the other members** in your group.

1. **Start** off by **working alone** to **plan** your personal learning timeline.

Think about the times in your life when you **learned a lot**.

There may be **life experiences** or events that caused you to **learn about** yourself as a person. There may be specific **projects of experiences** where you **learned a lot** technically.

- What happened?
- What did you learn?
- What about the **periods** in your life where you **didn't learn much**? Why was that?

2. **Draw** your personal learning timeline on a sheet of flip chart paper.

Think carefully about what you **are willing to share** with the **other members** of your group.

3. Now, with the other members of your group, **take turns to share your learning time line**. Describe the most important periods in your life for learning.



Exercise 2.3 (Personal Learning Timeline) 3 to 3

- **Hint**
- **This about reflecting on difficult or challenging periods of your life.** Think about the lessons you have learned and the skills you have acquired.
- For the timeline, draw peaks to represent **periods of intense learning**.
- The draw troughs to reflect time when you learned less. The peaks might (or might not) correspond to formal education.
- You might **have peaks** for a first unaccompanied camping trip with friends or organizing a big family gathering. You might have troughs for **boring periods** of your life.
- You can then share your timeline with the other members of your group.
- This will give you a framework for learning about each other and your previous experiences.
- You can use the timeline to learn about the life **histories** of the other team members.



Exercise 2.4 (Group Behavior Exercise) 1 to 3

We need to develop a set of ground rules about what sort of behaviors are acceptable in the group.

We want the other group members to **treat** us with respect. Right? But what does that mean? What is respectful behavior? Some members of your group might have very different expectations than you.

Things you consider **normal** may, in fact, be strange or even **offensive** to others in your group. You should try to find out if there are any **culturally sensitive areas** for any members of your group. You should also try to find out what has **annoyed** the other group members about working in teams in the past.

Work as a group to answer the following questions. Depending on the level of experience within the team, you might want to do some online information gathering to explore tools and techniques to answer the different questions.

Why not have each group member write a short report on one of the topics?



Exercise 2.4 (Group Behavior Exercise) 2 to 3

Use this exercise to learn new skills about ways of working in teams.

1. How will we communicate?
2. How will we collaborate?
3. How will we provide feedback?
4. How will we make decisions?
5. How will we handle conflict?
6. How will we prioritize work?
7. How will we measure our work?
8. How will we recognize or celebrate each other's contributions?

The outcome is a set of guidelines for methods of working, acceptable behavior and conduct within the group.

You should **list accepted behaviors** about how you would like to be treated by other members of the group.

You should also **list any unacceptable behaviors** you found out about in your discussions.



Exercise 2.4 (Group Behavior Exercise) 3 to 3

- **Hint**
- In groups working together for the first time, there could be different expectations about what constitutes (form) **acceptable behavior**.
- This is particularly important in groups with members from diverse cultural communities:
 - **Think** carefully about acceptable behaviors.
 - Discuss your previous experience of behavioral norms: what professional behaviors were accepted and what things did people do or say that caused controversy.
 - **Make a list** of unacceptable behaviors.
 - **Try to explore** areas where members of the group might be offended. This might include working practices, use of language and expectation around meeting attendance and timekeeping.
 - Finally, **make plans** for conflict resolution.
- It is better to consider tactics for making difficult decisions now while things are quite calm and relaxed.
- **Try to prepare a plan for decision-making** under **pressure** and when there are **heartfelt disagreements** within the group.



Exercise 2.5 (Creating Shared Goals) 1 to 2

In this exercise, you will try to **establish a sense of common purpose** in your group. A **well-developed shared mission** can help you perform better as a team.

1. **Work independently.** Each member of the group writes ‘what would you like this team to accomplish?’ Write a single accomplishment on each sticky note.
2. Now, **work as a group.** First, **identify common themes.** These are important; they are shared goals.
3. **Order the accomplishments** into priorities. Two categories is enough: high priority and low priority.
4. **Create a two-by-two matrix** (e.g. on a whiteboard). The two axes are shared vs. individual goals and low- vs. high-priority goals.
5. **Use the high-priority,** shared goals to create a mission statement for your team. Try to create a mission statement that is clear, simple and motivating.



Exercise 2.5 (Creating Shared Goals) 2 to 2

- **Hint**

Research shows that groups are **more effective** when they **have shared goals**. You might be able to increase the effectiveness of your group by understanding each other's goals.

Some group members may have very high expectations, in terms of work quality and effort. For other group members, it might be more a question of survival; 'let's just get through this'. You might not be able to change the expectations of other group members, but understanding their perspective might be useful.

Your **group performance** might **be improved** if you can **get consensus** on some aspects of group performance, such as quality of work. Some group members might be happy to accept producing more deliverables than others, if they feel everyone is committed to producing high-quality outputs.



Exercise 2.6 (Learning Journal) 1 to 4

- Reflect on the exercises you have completed, from Chapter 2.
- Make some notes in your learning journal.
- Think about what happened during each exercise. How did it go? What went well? What could have gone better?



Exercise 2.6 (Learning Journal) 2 to 4

Hint

- **Think** about how you performed in completing the exercises. Where they **easy**? Where they **challenging**? **Which** areas **need** some **further work**? Make some notes on these thoughts in your learning journal.
- Make some notes about the **quality of your solutions**. Try to **identify** areas where you **performed well** and those where you **might benefit** from **further reading** or other **learning**.
- If you are serious about becoming a professional practitioner in the software domain, you need to **use** your **learning journal** (or professional diary) to **record** your **activities**. The act of writing things down helps you think more clearly about what has happened. As you think more clearly, because of your writing, so you learn more about the **skills you are acquiring**.



Exercise 2.6 (Learning Journal) 3 to 4

- Don't just write a sequence of events; try to reflect on the skills you used. The main purpose is not only to create a timeline (although that can be useful to you later). The purpose is to **think** about the **successes** and **failure** of **what happened**. Try to **list** things that went well. Also **write** things that did not go so well. Then, make notes about what you might do differently next time.
- **Is your learning journal low tech or high tech?** I don't care. You could use a cloud-hosted software tool accessible from any device (like Evernote or Trello). Or, you could use a computer office application such as a word processor. Or, you could buy an expensive pen and nice notebook. Whatever works for you. I don't care. But, **write a journal**.
- Is your learning journal **personal** or **team based**? Both! Some of your learning will be **personal**. **Private. Confidential**, even. You need to think about your own reaction to the things you are learning. How did you feel during the exercise? What made you happy or uncomfortable? Why was that?
- Face it, learning is stressful, sometimes. If you don't challenge your own boundaries, you're not really learning anything new. Some, perhaps, more friendly, sociable people find working with others easy.



Exercise 2.6 (Learning Journal) 4 to 4

- You will want to keep this relationship **stuff private**, partly because some of your learning will be about working with others, and that **needs to be kept to yourself**. If you don't **write** something that should be **private**, then you're not *really* thinking about your own learning, or you don't yet have a good enough understanding of the boundaries between you and others.
- But you also need to learn, as a team, together. You need a group **Kanban board (Trello)**, **wiki site** or development platform, such as **GitHub**, to record your group decisions. You might like to use an instant messaging and file-sharing platform like **Slack**. These tools allow you to record group
- **decisions and exchange information with each other**. Kanban boards provide a visual overview of the project status. Using a shared wiki platform you can share architecture and design models. Instant messaging allows you to ask, and answer, questions.



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END OF LECTURE 02

Self-Organizing Teams (Exercises)

NEXT



Lecture 03: Agile Roles